Changkai Chen

List of Publications

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Peer-reviewed publications

Six first-author peer-reviewed articles and one preprint.

[1] Changkai Chen, Zhoujian Cao, and Jiliang Jing. Efficient and stable computation of gravitational-wave fluxes from generic Kerr orbits via a unified Heun-function framework. Physical Review D 114, 063002 (2026). DOI.

[2] Changkai Chen, Jiliang Jing, Zhoujian Cao, and Mengjie Wang. Complete quasinormal modes of type-D black holes. Physical Review D 112, 103036 (2025). DOI.

[3] Changkai Chen and Jiliang Jing. Gravitational wave fluxes on generic orbits in near-extreme Kerr spacetime: Higher spin and large eccentricity. Science China Physics, Mechanics & Astronomy 67, 110411 (2024). DOI.

[4] Changkai Chen and Jiliang Jing. Radiation fluxes of gravitational, electromagnetic, and scalar perturbations in type-D black holes: an exact approach. Journal of Cosmology and Astroparticle Physics 11 (2023) 070. DOI.

[5] Changkai Chen, Xiaohua Zhang, and Zhang Liu. A high-order compact finite difference scheme and precise integration method based on modified Hopf-Cole transformation for numerical simulation of n-dimensional Burgers' system. Applied Mathematics and Computation 372, 125009 (2020). DOI.

[6] Changkai Chen, Xiaohua Zhang, Zhang Liu, and Yage Zhang. A new high-order compact finite difference scheme based on precise integration method for the numerical simulation of parabolic equations. Advances in Difference Equations 2020, 15 (2020). DOI.

Preprint

[7] Changkai Chen, Xiaohua Zhang, Zhoujian Cao, Jiliang Jing, and Sheng Long. Complete total-transmission modes of Kerr black holes. arXiv:2609.39609 [gr-qc] (2026), preprint. arXiv.